University of Moratuwa

Department of Electronic and Telecommunication Engineering

EN3330 – Introduction to Engineering Optimization

Assignment 03

Lagrangian Duality, Saddle Points and Weak Duality

Name	Ranaweera R.P.D.D.H.
Index No.	220511N
Module	EN3330 – Introduction to Engineering Optimization
Lecturer	M. T. U. Sampath K. Perera
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Source code, figures and report:

https://github.com/Dulina14/Lagrangian-Duality-Saddle-Points-and-Weak-Duality

IN 22 EN3330 Assignment 03

M. T. U. Sampath K. Perera

1. Consider the constrained problem

$$\min_x x^2 \quad \text{subject to } x \geq 1. \tag{1}$$

Writing the constraint as $h(x) = 1 - x \leq 0$, the Lagrangian is

$$L(\mu, x) = x^2 + \mu(1 - x), \qquad \mu \geq 0. \tag{2}$$

(a) Plot $L(\mu, x)$ as a function of x for $\mu = 0, 1, 2, 3$ on the same axes. Mark the minimizer

$$x(\mu) = \arg\min_x L(\mu, x) \tag{3}$$

of each curve and describe how it moves as μ increases. [marks 30]

(b) Derive the dual function

$$d(\mu) = \min_x L(\mu, x) \tag{4}$$

analytically and plot it against μ. Identify

$$\mu^* = \arg\max_{\mu \geq 0} d(\mu) \tag{5}$$

and the dual optimal value d^*. [marks 30]

(c) Plot the primal function

$$g(x) = \max_{\mu \geq 0} L(\mu, x) \tag{6}$$

for $x \in [-1, 3]$. Show graphically that $g(x) = x^2$ for feasible x and $g(x) = +\infty$ for infeasible x. Hence explain why

$$\min_x g(x) \tag{7}$$

is equivalent to problem (1). [marks 40]

2. Consider the problem and Lagrangian in question 1. Let $x^* = 1$ and $\mu^* = 2$ be the optimal primal and dual solutions found there.

(a) Show that (x^*, μ^*) is a saddle point of L, i.e.

$$L(\mu, x^*) \leq L(\mu^*, x^*) \leq L(\mu^*, x) \quad \forall\, \mu \geq 0,\ \forall\, x. \tag{8}$$

Follow these steps:

(i) Compute $L(\mu^*, x^*)$.

(ii) Compute $L(\mu, x^*)$ and show that it does not depend on μ. Hence verify the left inequality.

(iii) Show that $L(\mu^*, x) = (x - 1)^2 + 1$. Hence verify the right inequality.

(iv) Plot $L(\mu, x^*)$ against μ and $L(\mu^*, x)$ against x (two small graphs) and mark the common value $L(\mu^*, x^*)$.

[marks 50]

1. Consider the constrained problem

$$\min_x x^2 \quad \text{subject to } x \geq 1. \tag{1}$$

Writing the constraint as $h(x) = 1 - x \leq 0$, the Lagrangian is

$$L(\mu, x) = x^2 + \mu(1 - x), \qquad \mu \geq 0. \tag{2}$$

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$$x(\mu) = \arg\min_x L(\mu, x) \tag{3}$$

of each curve and describe how it moves as μ increases. [marks 30]

Answer :-

$$L(\mu, x) = x^2 + \mu(1-x) \qquad \mu \geq 0$$

$$\frac{\partial L(\mu, x)}{\partial x} = 2x - \mu$$

$$\frac{\partial L(\mu, x)}{\partial x} = 0$$

$$0 = 2x - \mu$$

$$\frac{\mu^*}{2} = x^*$$

$$* \quad \frac{\partial^2 L(\mu, x)}{\partial x^2} = 2 > 0 \quad \therefore \text{At } \left(\mu^*, \frac{\mu^*}{2}\right) \text{ we have a minimum value.}$$

$$L\left(\mu^*, \frac{\mu^*}{2}\right) = \frac{\mu^{*2}}{4} + \mu^*\left(1 - \frac{\mu^*}{2}\right)$$

$$= \mu^* - \frac{\mu^{*2}}{4}$$

	$\mu = 0$	$\mu = 1$	$\mu = 2$	$\mu = 3$
$L(x, \mu)$	x^2	$x^2 - x + 1$	$x^2 - 2x + 2$	$x^2 - 3x + 3$
$x^*(\mu)$	0.00	0.50	1.00	1.50
minimum value	0.00	0.75	1.00	0.75
Allowed ? ($x \geq 1$)	No	No	Yes (on the boundary)	Yes (Away from the Optimum)

* The minimizer moves to the right. ($x(\mu) = \frac{\mu}{2}$, it moves to the right at a rate of 0.5 for each μ increment by 1.)

When $\mu = 0 \Rightarrow x = 0.0$ (not allowed)

$\mu = 1 \Rightarrow x = 0.5$ (")

$\mu = 2 \Rightarrow x = 1.0$ (Exactly on the boundary and the correct answer)

$\mu = 3 \Rightarrow x = 1.5$ (Allowed, but past the best point)

∴ When μ increases, the minimizer travel from the unconstrained answer ($x=0$) toward the constrained answer ($x=1$). It reaches the constrained answer at ($\mu^* = 2$).

(b) Derive the dual function

$$d(\mu) = \min_x L(\mu, x) \quad (4)$$

analytically and plot it against μ. Identify

$$\mu^* = \arg\max_{\mu \geq 0} d(\mu) \quad (5)$$

and the dual optimal value d^*. [marks 30]

(b) $d(\mu) = \min_x L(\mu, x)$

$= x^2 + \mu(1 - x)$

$d(\mu) = x^2 + \mu x + \mu$

$x^*(\mu) = \frac{\mu}{2}$

$d(\mu) = \mu - \frac{\mu^2}{4}$

$\frac{d[d(\mu)]}{d\mu} = 1 - \frac{\mu}{2} \Rightarrow 1 - \frac{\mu}{2} = 0 \Rightarrow \mu = 2$

$\frac{d^2[d(\mu)]}{d\mu} = -\frac{1}{2} < 0$ ∴ Maximum.

($d(\mu)$ is concave shaped.)

∴ $\mu^* = 2$, $d(2) = 1$

∴ The dual optimum equals the primal optimum.

Figures for Question 1(a) and 1(b)

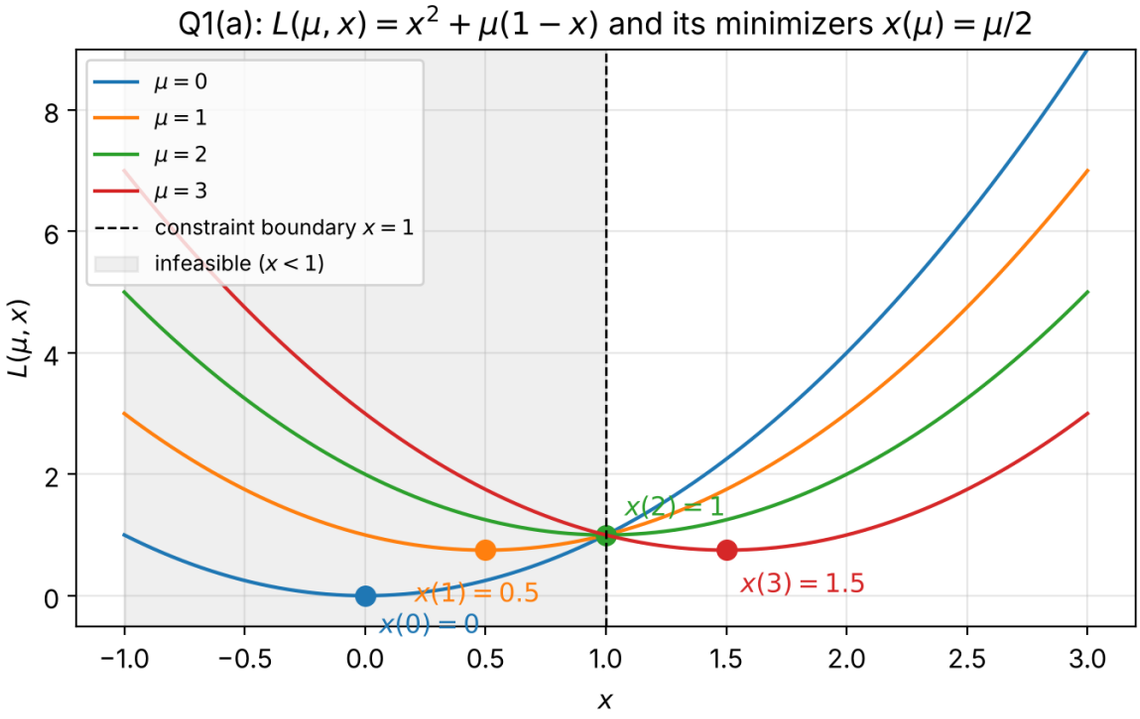


Figure 1: $L(\mu, x) = x^2 + \mu(1 - x)$ against x for $\mu = 0, 1, 2, 3$. The dots mark the minimiser $x(\mu) = \mu/2$ of each curve, which moves right as μ increases and reaches the constraint boundary $x = 1$ at $\mu = 2$. The shaded region ($x < 1$) is infeasible.

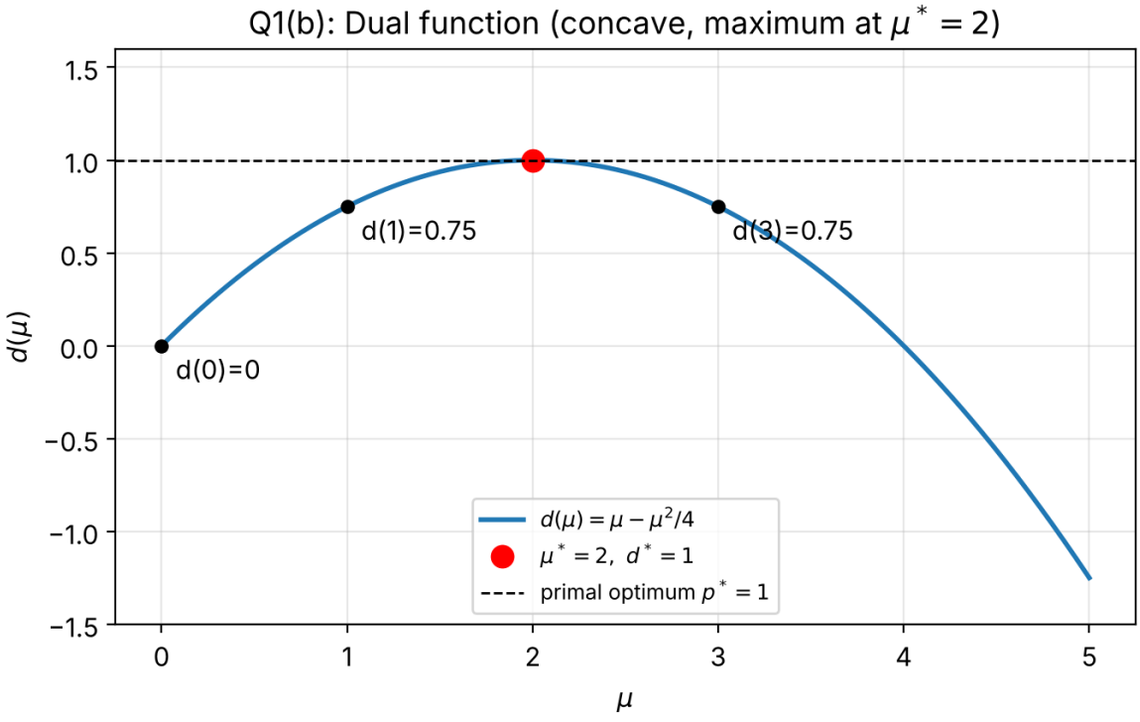


Figure 2: Dual function $d(\mu) = \mu - \mu^2/4$. It is concave, with its maximum at $\mu^* = 2$, where $d^* = 1$ equals the primal optimum $p^* = 1$ (dashed line), showing zero duality gap.

(c) Plot the primal function

$$g(x) = \max_{\mu \geq 0} L(\mu, x) \tag{6}$$

for $x \in [-1, 3]$. Show graphically that $g(x) = x^2$ for feasible x and $g(x) = +\infty$ for infeasible x. Hence explain why

$$\min_x g(x) \tag{7}$$

is equivalent to problem (1). [marks 40]

(c) $g(x) = \max_{\mu \geq 0} L(\mu, x)$

$= \max_{\mu \geq 0} x^2 + \mu(1-x)$

for fixed x value $x^2 + \mu(1-x)$ is a line.

Case 01.

$(1-x) \leq 0$ (according to the constraint.)

line is the flat or decreasing further for $\mu \geq 0$

$\therefore x^2 = \max_{\mu \geq 0} (x^2 + \mu(1-x))$

Case 02.

$(1-x) > 0$ (break the constraint)

line is increasing for $\mu \geq 0$

$\therefore +\infty = \max_{\mu \geq 0} (x^2 + \mu(1-x))$

$$g(x) = \begin{cases} x^2 & ; \text{ feasible } x \\ +\infty & ; \text{ otherwise} \end{cases}$$

$$\min_x g(x) = \min_x \max_{\mu \geq 0} L(\mu, x) = \begin{cases} x^2 & ; \text{ feasible } x \ (x \geq 1) \\ +\infty & ; \text{ otherwise } (x < 1) \end{cases}$$

$\therefore \min_x g(x)$ can never be on the infeasible region.

In the feasible region $g(x)$ is x^2, and $\min_x x^2$ this in $x \geq 1$ region is exactly the same problem in (1).

Figure for Question 1(c)

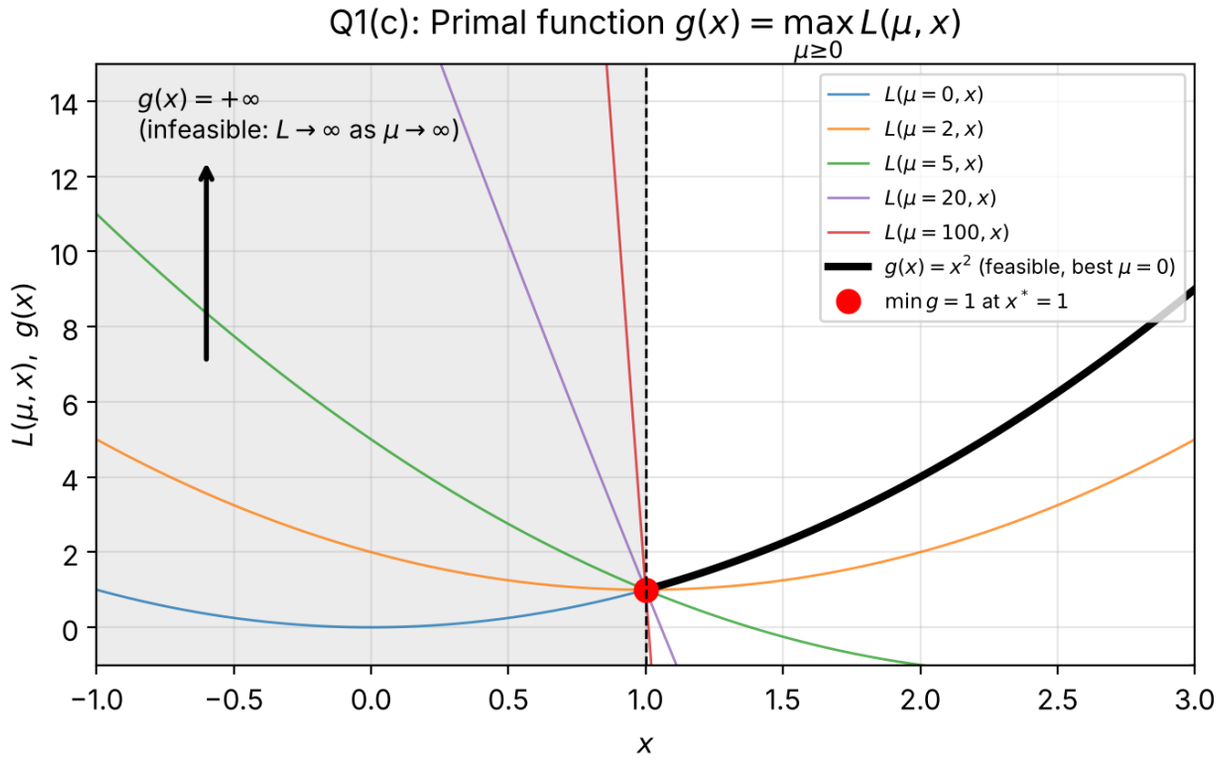


Figure 3: $L(\mu, x)$ against x for increasing $\mu = 0, 2, 5, 20, 100$. For infeasible $x < 1$ (shaded) the curves grow without limit as μ increases, so $g(x) = +\infty$. For feasible $x \geq 1$ the highest curve is the $\mu = 0$ curve, so $g(x) = x^2$ (thick black line). The minimum of g is 1 at $x^* = 1$, the same as problem (1).

2. Consider the problem and Lagrangian in question 1. Let $x^* = 1$ and $\mu^* = 2$ be the optimal primal and dual solutions found there.

(a) Show that (x^*, μ^*) is a saddle point of L, i.e.

$$L(\mu, x^*) \le L(\mu^*, x^*) \le L(\mu^*, x) \quad \forall \mu \ge 0, \forall x. \tag{8}$$

Follow these steps:

(i) Compute $L(\mu^*, x^*)$.

(ii) Compute $L(\mu, x^*)$ and show that it does not depend on μ. Hence verify the left inequality.

(iii) Show that $L(\mu^*, x) = (x-1)^2 + 1$. Hence verify the right inequality.

(iv) Plot $L(\mu, x^*)$ against μ and $L(\mu^*, x)$ against x (two small graphs) and mark the common value $L(\mu^*, x^*)$.

[marks 50]

(a) (i) $L(\mu, x) = x^2 + \mu(1-x)$

$L(\mu^*, x^*) = L(2, 1) = 1 + 2(1-1) = 1$

(ii) $L(\mu, x^*) = L(\mu, 1) = 1 + \mu(1-1) = 1$

μ is not remain in the expression. $\therefore$ $L(\mu, x^*)$ does not depend on μ.

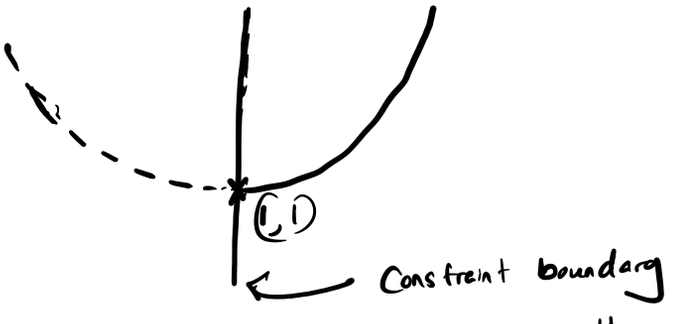
$x^* = 1$ is on the boundary. $\therefore$ $h(x^*) = 0$,

$\mu h(x^*) = 0$, penalty term doesn't affect.

$\Rightarrow$ $1 = L(\mu, x^*) \le L(\mu^*, x^*) = 1$

Left inequality is satisfied.

(iii) $L(\mu^*, x) = x^2 - 2x + 2 = (x-1)^2 + 1$



$L(\mu^*, x)$ is a convex function, with the minimum $x = 1$ which is also the boundary for the constraint.

$\therefore$ $L(\mu^*, x) \ge 1$ for $x \ge 1$

$\Rightarrow 1 = L(\mu^*, x^*) \le L(\mu^*, x)$ $\therefore$ Holds the inequality right side.

Figure for Question 2(a)(iv)

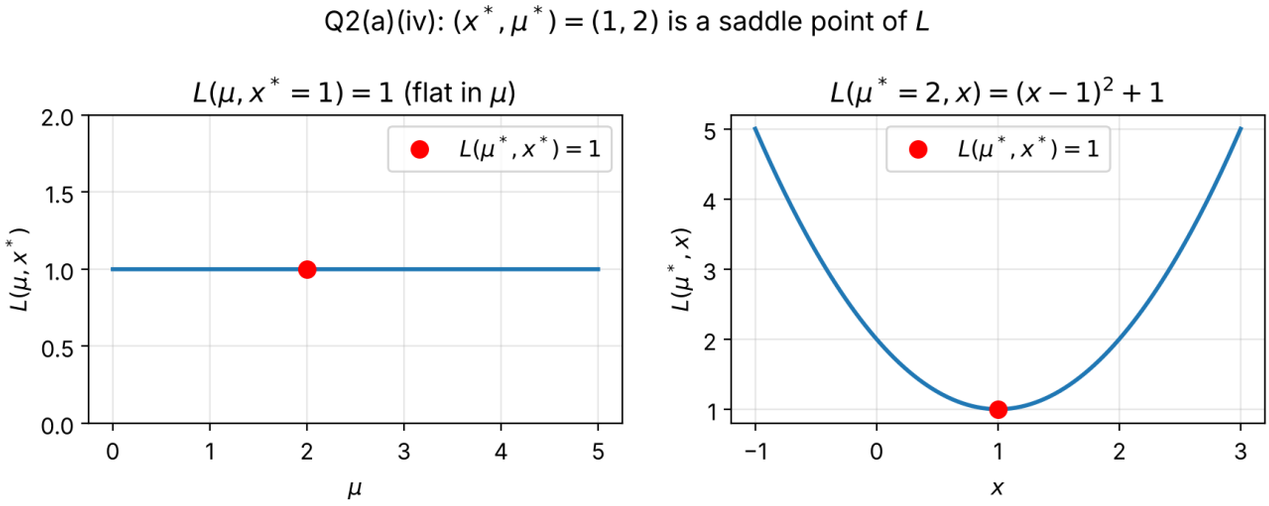


Figure 4: Saddle point check. Left: $L(\mu, x^*) = 1$ for every μ (flat line), so no μ rises above $L(\mu^*, x^*)$ (left inequality). Right: $L(\mu^*, x) = (x-1)^2 + 1$ has its lowest value at $x = 1$, so no x goes below it (right inequality). The red dots mark the common value $L(\mu^*, x^*) = 1$.

(b) The optimality argument for the right inequality can be written as

$$f(x^*) \overset{(1)}{=} L(\mu^*, x^*) \overset{(2)}{\leq} L(\mu^*, x) \overset{(3)}{=} f(x) + \mu^* h(x) \overset{(4)}{\leq} f(x), \tag{9}$$

for any feasible x. Give a one-line reason for each step (1)–(4). What does the left inequality tell you about $h(x^*)$ if $L(\mu, x^*)$ must stay finite as $\mu \to \infty$?

[marks 25]

(c) Verify that the complementary slackness condition

$$\mu^* h(x^*) = 0 \tag{10}$$

holds. Now change the constraint to $x \geq -1$. Is the unconstrained minimizer $x = 0$ of x^2 feasible? Is the constraint active there? What value would you expect for μ^*, and why?

[marks 25]

3. **Weak duality.** Consider the general problem

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) \quad \text{subject to } h(\boldsymbol{x}) \leq 0, \tag{11}$$

with Lagrangian $L(\mu, \boldsymbol{x}) = f(\boldsymbol{x}) + \mu h(\boldsymbol{x})$ and dual function

$$d(\mu) = \min_{\boldsymbol{x}} L(\mu, \boldsymbol{x}), \qquad \mu \geq 0. \tag{12}$$

Let p^* denote the optimal value of problem (11).

(a) Let $\mu \geq 0$ and let $\boldsymbol{x}$ be any feasible point (so $h(\boldsymbol{x}) \leq 0$). Give the reason for each of the two inequalities in

$$d(\mu) \overset{(1)}{\leq} L(\mu, \boldsymbol{x}) = f(\boldsymbol{x}) + \mu h(\boldsymbol{x}) \overset{(2)}{\leq} f(\boldsymbol{x}). \tag{13}$$

Conclude that $d(\mu) \leq p^*$ for every $\mu \geq 0$. Then check this numerically for question 1: using $d(\mu)$ from Q1(b), verify that $d(\mu) \leq p^* = 1$ for $\mu = 0, 1, 2, 3$. [marks 30]

(b) For question 1, sketch $L(\mu, x) = x^2 + \mu(1 - x)$ as a function of μ for $x = 0, 1, 2$ on the same axes (these are straight lines). Draw the lower envelope of the three lines, which approximates $d(\mu)$. Is it concave or convex? Use this picture to explain in one or two sentences why $d(\mu)$ is concave in μ for *any* f, even a non-convex one. [marks 20]

(b) The optimality argument for the right inequality can be written as

$$f(x^*) \overset{(1)}{=} L(\mu^*, x^*) \overset{(2)}{\leq} L(\mu^*, x) \overset{(3)}{=} f(x) + \mu^* h(x) \overset{(4)}{\leq} f(x), \tag{9}$$

for any feasible x. Give a one-line reason for each step (1)–(4). What does the left inequality tell you about $h(x^*)$ if $L(\mu, x^*)$ must stay finite as $\mu \to \infty$?

(b) (1) $L(\mu^*, x^*) = f(x^*)$

$\because L(\mu^*, x^*) = f(x^*) + \mu^* h(x^*)$, $h(x^*) = 0$ since x^* on the boundary of the constraint.

(2) $L(\mu^*, x^*) \leq L(\mu^*, x)$

This is same as what we proved in 2 (a) (iii). i.e. $(x-1)^2 + 1 \geq 1$

(3) $L(\mu^*, x) = f(x) + \mu^* h(x)$

Definition of Lagrangian, evaluated at x.

(4) $h(x) \leq 0$ and $\mu^* \geq 0$, $\therefore \mu h(x) \leq 0$

$\Rightarrow f(x) + \mu h(x) \leq f(x)$

$(\because \leq 0)$

$L(\mu, x^*) \leq L(\mu^*, x^*)$

$f(x^*) + \mu h(x^*) \leq L(\mu^*, x^*)$

If we think this $h(x^*)\mu + f(x^*)$ as a line,

$L(\mu^*, x^*)$ becomes an upper bound.

$h(x^*)$	what happen when $\mu \to \infty$ to $h(x^*)\mu + f(x^*)$	Does the inequality holds
(>0)	grows to $(\to) + \infty$	No
$(=0)$	$f(x^*)$	Yes
(<0)	goes to $(\to) -\infty$	Yes

$\therefore h(x) \leq 0 \Rightarrow$ Obey to the constraint to the inequality to hold

(c) Verify that the complementary slackness condition

$$\mu^* h(x^*) = 0 \qquad (10)$$

holds. Now change the constraint to $x \geq -1$. Is the unconstrained minimizer $x = 0$ of x^2 feasible? Is the constraint active there? What value would you expect for μ^*, and why?
[marks 25]

$\mu^* \, h(x^*) = 0$

$\Rightarrow$ At least one of the terms equel to zero.

$h(x^*) = 0$, $x^* = 1$

The solution is on the boundary. (Constraint is active)

$\mu^* = 2$ $\qquad h(x^*) = 1 - x^* = 1 - 1 = 0$

$\mu^* \, h(x^*) = 2 \times 0 = 0$

* Complementary slackness holds.

$x \geqslant -1$

$-x - 1 \leqslant 0$

1. Unconstreint minimizer is $x = 0 \geqslant -1$ $\therefore$ Yes, Unconstraint minimzer is feasible.

$h(0) = -1 \leqslant 0$

2. "No Constrain t is not active.
Constraint to be active $h(0) = 0$, but here $h(0) = -1$

3. $\mu^* = 0$

Complementary slackness to be held $\mu^* \, h(x^*) = 0$, since $h(x^*) \neq 0$

$\therefore$ $\mu^* = 0$.

3. **Weak duality.** Consider the general problem

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with Lagrangian $L(\mu, \boldsymbol{x}) = f(\boldsymbol{x}) + \mu h(\boldsymbol{x})$ and dual function

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Let p^* denote the optimal value of problem (11).

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$$d(\mu) \overset{(1)}{\le} L(\mu, \boldsymbol{x}) = f(\boldsymbol{x}) + \mu h(\boldsymbol{x}) \overset{(2)}{\le} f(\boldsymbol{x}). \tag{13}$$

Conclude that $d(\mu) \le p^*$ for every $\mu \ge 0$. Then check this numerically for question 1: using $d(\mu)$ from Q1(b), verify that $d(\mu) \le p^* = 1$ for $\mu = 0, 1, 2, 3$. [marks 30]

(a) $d(\mu^{(1)}) \le L(\mu, x)$

(1) $\min_{x} L(\mu, x) \le L(\mu, x)$

lower bound of $L(\mu, x)$ ∴ Inequality holds.

(2) $f(x) + \mu h(x) \le f(x)$

$\Rightarrow \mu \ge 0$ & $h(x) \le 0 \Rightarrow \mu h(x) \le 0$ (∵ x is feasible)

$\Rightarrow f(x) + \mu h(x) \le f(x)$

≤ 0

Adding a negative value to $f(x)$, always makes it lesser.

∴ for every feasible ∴ x^* also adhere to that.

$f(x^*) = p^*$

$d(\mu) \le f(x^*) = p^*$

$d(\mu) \le p^*$ for every $\mu \ge 0$

Q1

μ	$d(\mu) = \mu - \frac{\mu^2}{4}$	Is $d(\mu) \leq 1 = p^*$
0	$0 - 0 = 0.00$	Yes.
1	$1 - 0.25 = 0.75$	Yes.
2	$2 - 1 = 1.00$	Yes
3	$3 - 2.25 = 0.75$	Yes.

(b)

For a fixed x, the Lagrangian $L(\mu, x) = x^2 + \mu(1 - x)$ is a straight line in μ, with intercept x^2 and slope $(1 - x)$:

x	L(μ, x)	Intercept	Slope
0	μ	0	+1 (rising)
1	1	1	0 (flat)
2	4 − μ	4	−1 (falling)

The lines for $x = 0$ and $x = 1$ cross at $\mu = 1$, and the lines for $x = 1$ and $x = 2$ cross at $\mu = 3$. Taking the lowest line at each μ gives the lower envelope (thick black line in Figure 5):

$E(\mu) = \mu$ for $0 \leq \mu \leq 1$, $E(\mu) = 1$ for $1 \leq \mu \leq 3$, $E(\mu) = 4 - \mu$ for $\mu \geq 3$.

Concave or convex? The envelope rises, stays flat and then falls, giving a “tent” (hill) shape, so it is **concave**. It lies on or above the true dual function $d(\mu) = \mu - \mu^2/4$ (red dashed curve), because it uses only three values of x instead of minimising over all x. Each line touches $d(\mu)$ at one point, $\mu = 2x$ (at $\mu = 0$, 2 and 4), so adding more values of x would make the envelope approach $d(\mu)$.

Why $d(\mu)$ is concave for any f: for each fixed x, $f(x)$ and $h(x)$ are just numbers, so $L(\mu, x) = f(x) + \mu h(x)$ is always a straight line in μ, however complicated or non-convex f is. Since $d(\mu)$ is the pointwise minimum (lower envelope) of these lines, and the minimum of any set of straight lines can only bend downwards, $d(\mu)$ is always concave.

Interpretation: because $d(\mu)$ is concave, any maximum found is the global maximum, so the dual problem is always a well-behaved problem, and by part (a) its value always gives a valid lower bound on p^*.

Figure for Question 3(b)

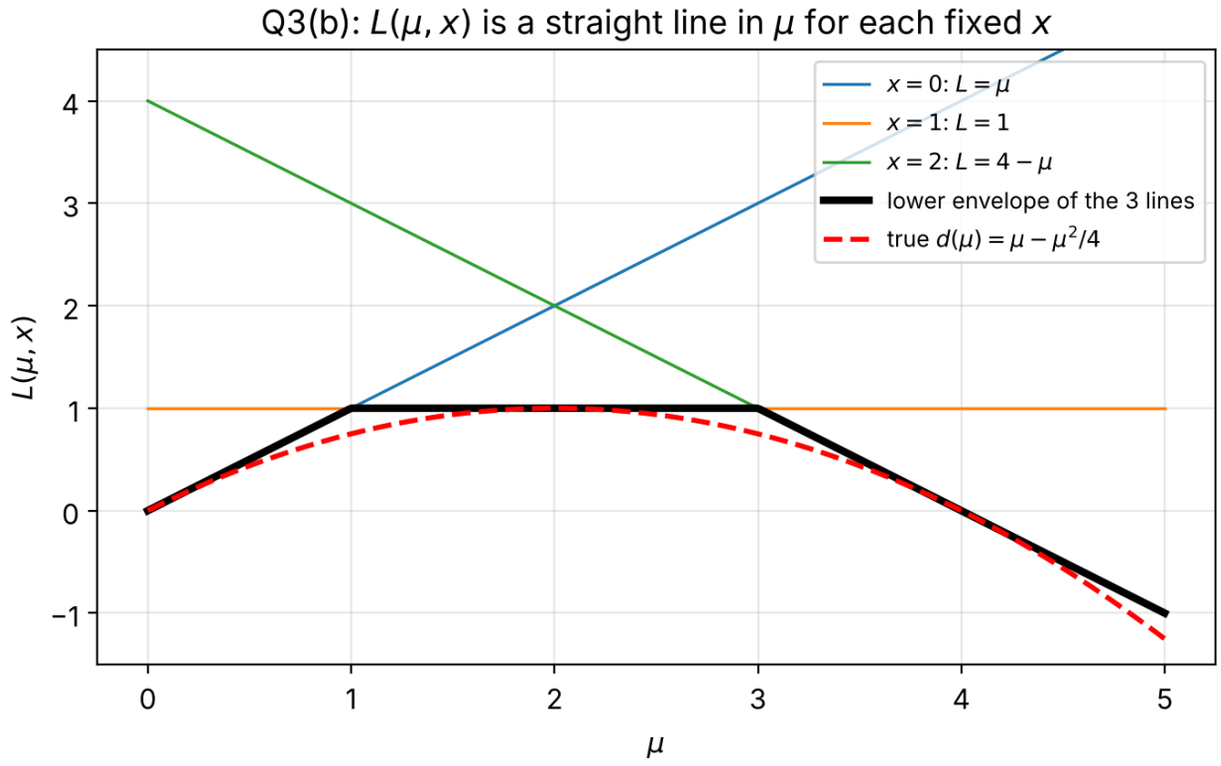


Figure 5: $L(\mu, x)$ as a function of μ for $x = 0, 1, 2$ (straight lines). The thick black line is their lower envelope, which is concave (tent-shaped) and lies on or above the true dual function $d(\mu) = \mu - \mu^2/4$ (red dashed curve).

Submission

- Upload a report as a pdf file named as "Yourindexno_EN3330_A03.pdf". Include the index number and the name within the report as well.
- The interpretation of results and the discussion are important in the report.
- Pay careful attention to formatting such as font size, spacing, and margins.
- Include a title page with necessary information (e.g., title, author, date, index no).
- Use consistent and professional formatting throughout the document.
- Plagiarism will be checked and in cases of plagiarism, an extra penalty of 50% will be applied. In case of copying from each other, both parties involved will receive a grade of zero for the assignment. Academic integrity is of utmost importance, and any form of plagiarism[1] or cheating will not be tolerated.
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[1] https://en.wikipedia.org/wiki/Plagiarism